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Addressable CMOS Image Sensor Array

Abstract

This work presents the design and verification of a fully integrated 32×32 CMOS image sensor mixed-signal system. The sensor integrates a 4T pixel array, row/column addressing, level shifting, correlated double sampling (CDS) for noise reduction, and an 8-bit successive approximation register (SAR) analog-to-digital converter. The design emphasizes system-level integration, highlighting how pixel-level analog behavior, biasing strategies, noise sources, and timing constraints interact when multiple subsystems are combined. A hierarchical design and verification methodology is used to validate each block independently before integration, enabling isolation of non-ideal effects such as limited pixel swing, charge injection, and timing sensitivity in the SAR logic. Simulation results demonstrate correct pixel readout, CDS noise suppression, and reliable SAR conversion with moderate resolution and speed suitable for low-resolution imaging applications. Beyond achieving functional imaging front-end operation, this project provides practical insight into real-world analog IC design tradeoffs, illustrating the challenges of coordinating analog settling, digital control, and mixed-voltage operation within a realistic CMOS image sensor architecture.

Introduction

CMOS image sensors integrate photodetectors, analog amplification, and signal-processing, enabling compact, low-power, and scalable imaging systems. From an analog IC design perspective, image sensors present a rich design space where small-signal analog performance, noise behavior, device matching, biasing, and layout directly affect system-level performance, making them a valuable platform for studying mixed-signal integration.

The motivation for this project is to design an addressable CMOS image sensor as a comprehensive capstone in analog integrated circuit design. Unlike isolated circuit blocks, an image sensor requires close coordination between pixel-level analog circuits, switching and addressing logic, and mixed-signal readout circuitry. This project provides practical experience applying core analog design concepts such as biasing, transistor sizing, noise optimization, and parasitic management within a realistic, end-to-end system that interfaces with a physical input: light intensity.

The 32×32 CMOS image sensor array developed in this project represents a simplified imaging front end applicable to low-resolution sensing tasks such as light intensity mapping and embedded vision. More importantly, it mirrors industry-relevant analog IC design workflows, where device-level optimization, circuit robustness, and system-level integration must be addressed simultaneously to achieve a functional and reliable mixed-signal integrated system.

Overview

This project implements an addressable CMOS image sensor system that integrates a pixel array, addressing logic, analog signal conditioning, and analog-to-digital conversion. The system is designed to sequentially access pixels within a 2D array, condition the resulting analog

signals, and convert them into digital outputs suitable for further processing or analysis. A high-level system block diagram is shown in Figure 1.

At the core of the design is a 32x32 active pixel sensor (APS) array operating on a 3.3 V analog supply. Pixel access is controlled by row and column decoders implemented on a 1.0 V digital domain to reduce power consumption. A level shifter translates the 32-bit decoder outputs from 1.0 V to 3.3 V control signals required by the pixel array, enabling reliable mixed-voltage operation while minimizing digital noise coupling.

During readout, the selected pixel's analog output is processed by a correlated double sampling (CDS) buffer to suppress pixel thermal and flicker noise and reduce fixed-pattern noise. The signal is then digitized by an 8-bit successive approximation register (SAR) ADC. The SAR architecture is selected for its balance of resolution, power efficiency, and ease of integration in CMOS technology. ADC conversion is synchronized with pixel addressing and CDS timing, enabling sequential digitization of pixel values as the array is scanned.

We used a hierarchical design and verification method to manage system complexity and ensure correct operation at each level of integration. Each functional block is designed and validated independently, followed by incremental integration of the decoder, level shifter, pixel array, CDS buffer, and ADC, before simulating full-system operation with all blocks integrated and operating under coordinated control and timing. This approach ensures robust system functionality and provides insight into the contribution of each block to overall performance.

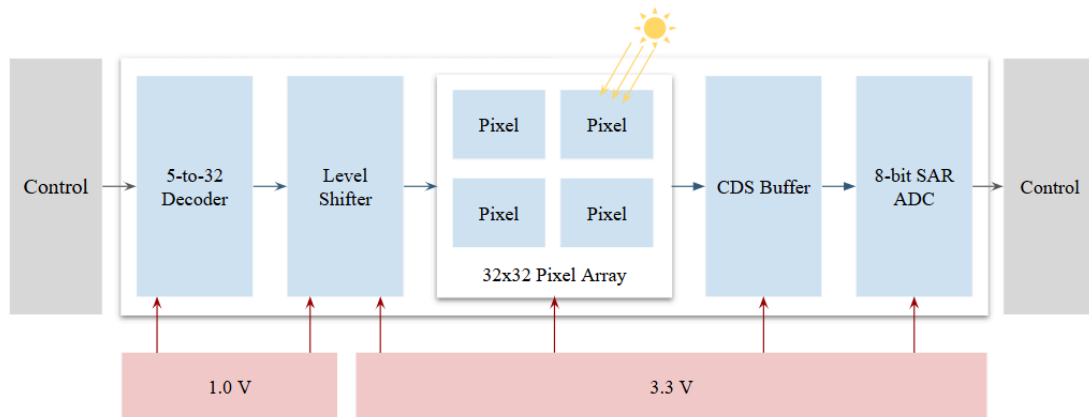


Figure 1. High-level block diagram

Circuit Designs, Analysis + Theory, and Design Verification

Pixel Cell

Design (Refer to Appendix C for Schematic & Layout of the pixels)

We chose a standard 4T active pixel sensor architecture because it is one of the most stable and widely used pixel designs in CMOS imaging. Compared to simpler pixels, the 4T structure provides a cleaner and more controlled readout path: the pixel signal is transferred to a floating diffusion node and then buffered by a source follower before going onto the column line. This isolation makes the output more repeatable from frame to frame and improves image quality, since the readout is less sensitive to disturbances on the column bus.

We could size the reset transistor longer to reduce leakage and improve reset robustness, which helps keep the FD node well-defined and reduces unwanted discharge effects between operations. We also sized the source follower wider to increase drive strength and improve

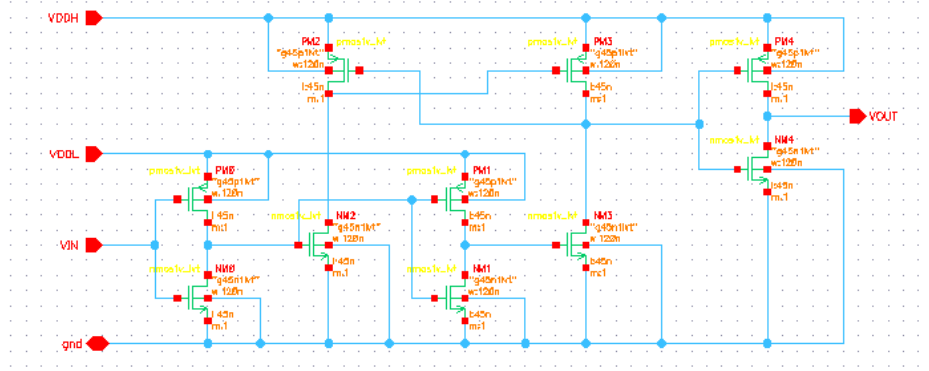


Figure 4. Level shifter 1-bit cell schematic

Correlated Double Sampling (CDS) Buffer Design

The CDS buffer circuit is a switched-capacitor circuit that subtracts two sampled voltage values to remove noise and offsets. The OTA's output is shorted to its input through feedback capacitors, which sets them as unity-gain buffers. In the first phase, the pixel reset voltage is sampled onto C_1 , which contains the pixel's thermal and fixed-pattern noise (FPN), and the OTA's offset. In the second phase, the pixel signal voltage is sampled and subtracted from the reset level. The charge on C_1 passes on to C_2 , which cancels out the OTA's offset. The gain of the buffer is set by the capacitor ratios, and since we want the column voltage level to stay around the same as the pixel output, we set all capacitors to be equal for unity gain.

$$V_{out} = \frac{C_2}{C_F} \left(V_{in} \frac{C_1}{C_F} \right)$$

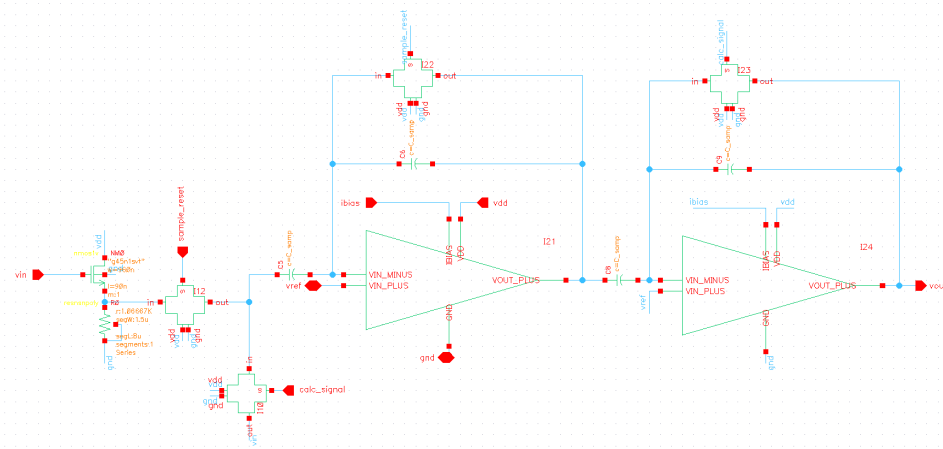


Figure 5. CDS Buffer Schematic

Transmission gates are used for the switches because they can pass the full logic voltage range 0 to Vdd without degradation, and have a lower and less signal-dependent R_{on} , which provides higher linearity. The amplifier used was a modified version of the lab 3 amp to make it single-ended with increased transistor sizing to optimize for noise. This OTA was chosen because it has a high UGBW of $\sim 250\text{MHz}$, so it has a fast settling time, which is required for accurate switching and sampling.

A source follower buffer was added between the input and the first switch to prevent back injection. This occurs because the toggling of the switch's gate voltage pushes charge into the source and drain, and since the column output is high impedance, this makes it especially sensitive to shifts in voltage. The transistor in the SF buffer was made wide to give much lower output impedance and larger output capacitance to absorb this charge. Schematics and layouts can be found in [Appendix B](#).

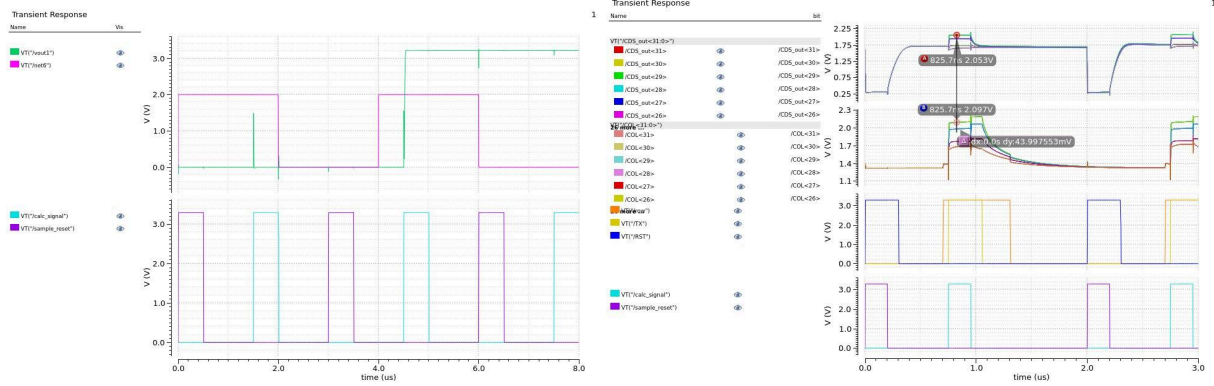


Figure 6. CDS Simulation Results

Verification

The simulation results show that in the first half of v_{in} , the voltages at the sample reset phase and the calc signal phase are the same, so the CDS buffer outputs around 0V. In the second half, the sample reset phase samples 0V while the calc signal phase samples 2V, and the difference is reflected in the CDS output. The simulation with the pixel used as the input shows the CDS providing a relatively stable voltage to feed into the ADC, which is slightly lower than the pixel column output due to the removal of noise and offsets.

The input referred noise of the CDS buffer was measured to be approximately $63\mu\text{V rms}$. Since CDS cancels out pixel-level noise, one of the important aspects was to lower the noise of the CDS itself. From the noise summary ([Appendix B](#)), it can be seen that the flicker noise of the NM0 SF transistor was dominating, so we reduced it by increasing the length of the SF transistor. The power consumption was measured to be 1.72mW per CDS buffer. This is much higher than typical per-column CDS buffer power usage, and is likely due to our higher 3.3V supply voltage, high bias current for the op amps for faster settling, and increased input transistor widths for noise reduction.

Successive Approximation Register Analog to Digital Converter (SAR ADC)

Design

An ADC is necessary to convert the analog signal from the CDS buffer into a digital output that can be used to further process images sensed by the pixel array. To do so, our 8-hit SAR ADC uses three major subcomponents: a sample/hold circuit that also functions as a digital-to-analog converter (DAC), a synchronous comparator, and successive approximation register digital logic. The connections between these subcomponents are depicted in the schematic below:

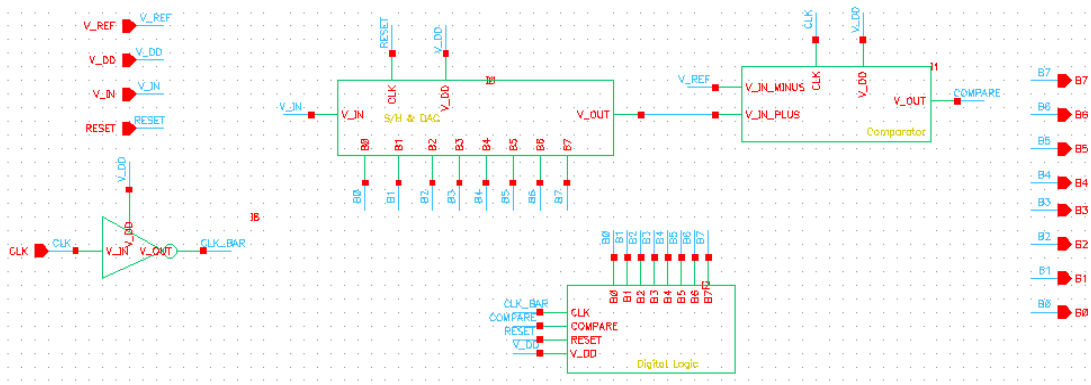


Figure 7. Top-level circuit diagram for SAR ADC.

The sample/hold DAC circuit consists of a transmission gate and a capacitor/inverter array, shown in Figure 8. The transmission gate “samples” the CDS buffer output by turning on for a certain pulse duration at the desired sampling frequency. Our sample/hold DAC consists of an array of eight inverters and capacitors, corresponding to the bit resolution of our ADC. These capacitors and inverters are sized according to their bit significance. For example, the capacitor connected to the most significant bit B7 is sized up to $2^7 C_0 = 128 C_0$, where C_0 (1.995 fF) is the minimal unit capacitance for MIM capacitors in gpdk045. It is important to note that for a capacitive DAC, it is the ratio between successive capacitors that matters. Meanwhile, the capacitor connected to the least significant bit B0 is sized to $2^0 C_0 = C_0$. The minimal unit capacitance 1.995 fF was retained as C_0 to maximize the speed of DAC operation and to minimize die area. When the transmission gate is closed, all bits charge up to “1”. When the transmission gate is open, DAC conversion occurs, and the sample/hold DAC output obtains the following value:

$$V_x = V_{in} + \sum_{n=0}^{N-1} \frac{C_o 2^n}{C_o 2^N} \overline{B_n}$$

Equation 1. Sample/hold DAC output voltage formula. B_n represents the output bit from the SAR digital logic. N represents the total number of output bits for the ADC.

The synchronous comparator, a differential amplifier with positive feedback (suggested by Professor Molnar in his ECE 5540 slides), compares V_{in}^- to V_{in}^+ when clocked. Positive feedback increases the speed of comparator operation. Since our design is differential yet a single-ended output was needed, V_{out}^- is not utilized while V_{out}^+ is fed into the SAR digital logic. Sweeping various transistor sizes, we achieved optimal performance when the common mode source degeneration transistor was sized to a width of 4.8 μm and a length of 45 nm while all other transistors were sized to a width of 2.4 μm and a length of 45 nm. When the clock signal is low, the internal nodes V_{int}^- and V_{int}^+ are reset to V_{DD} , thus turning off the input differential pair. When the clock signal is high, the internal nodes are “released”, biasing the differential bias at a certain current I_{bias} . The input voltages generate a current difference ΔI between the two “halves” of the differential amplifier. The internal nodes then slew down at the rate described in [Appendix D](#). Thus, when the internal node voltages are less than $V_{DD} - V_{TH}$, current is injected into the opposite nodes via the gates, amplifying the difference between V_{in}^- and V_{in}^+ . Our design is depicted in Figure 9.

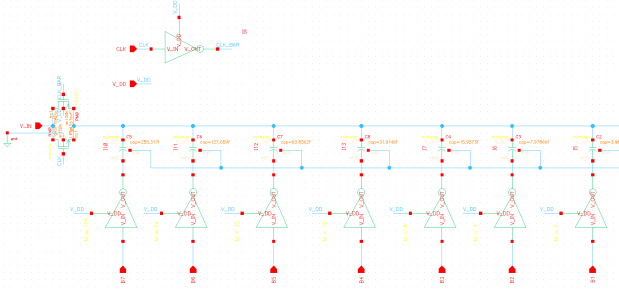


Figure 8. Sample/hold DAC schematic.

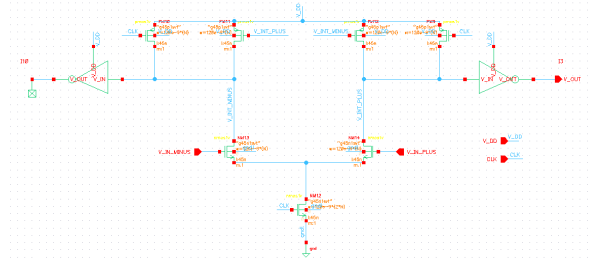


Figure 9. Comparator schematic.

The SAR digital logic “reads” the comparator output to digitally approximate the analog input to the ADC. As the pixels output four distinct voltage values, we designed an 8-bit ADC for moderate accuracy and speed. Our design is depicted in [Appendix D](#). Using negative clock edge-triggered D flip-flops (setup time = 40 ps, hold time = -5 ps), we created eight pointers (bottom row of flip-flops) and eight registers (top row of flip-flops). When the RESET signal toggles high, the leftmost pointer outputs a “high” signal, which then sets the leftmost register to also output a “high” signal. Upon the falling edge of RESET, the comparator output is read through B7. Upon the next falling clock edge, the “high” signal from the leftmost pointer is shifted right and sets the second leftmost register to also output a “high” signal. Upon the falling edge of this signal (the next falling clock edge), the comparator output is read through B6. This pattern is repeated until all eight bits have “read out” the comparator output. Essentially, after sampling and holding, each bit is computed in succession over 8 clock cycles, starting from the most significant bit.

Verification

Sample/hold DAC functionality was verified via operation within the rest of the ADC (i.e. in conjunction with the comparator and SAR digital logic). The input voltage was linearly swept from 0-2.5 V over a period of 290 μ s, followed by a 10 μ s drop back to 0 V. The clock signal was set to oscillate between 0 and V_{DD} (2.5 V) at $\sim$ 143 kHz. The maximum input voltage value was chosen based on the maximum output voltage of the CDS buffer. As expected, when the RESET signal (used as the clock input to the sample/hold DAC) is toggled, the output matches the input. When the RESET signal is low, the sample/hold DAC approximates a constant value regardless of the input voltage. This is expected since the output bits from the SAR digital logic are inverted before being fed into the capacitor array. These results are depicted in Figure 10 below.

Comparator functionality was verified by linearly sweeping V_{in}^+ from 0 V to $V_{DD} = 2.5$ V. V_{in}^- was set to a reference voltage $V_{REF} = V_{DD}/2 = 1.25$ V that V_{in}^+ was compared two with each clock cycle. As expected, the comparator output remains at 0 V when $V_{in}^+ < V_{REF}$. When $V_{in}^+ > V_{REF}$, the comparator output voltage = V_{DD} when the clock signal is high. These results are depicted in [Appendix D](#).

The SAR logic was tested while integrated with the rest of the ADC, sweeping the input voltage from 0 V to 2.5 V over 290 μ s, followed by a 10 μ s drop back to 0 V. After sweeping various clock periods, optimal performance and accuracy was achieved at 7 μ s ($\sim$ 143 kHz). To ensure accurate 8-bit resolution, RESET was pulsed every nine clock cycles (eight cycles for input approximation + one cycle for output settling). Thus, it takes $\sim$ 63 μ s to digitally approximate one sampled input voltage value. These results are shown in Figure 11 below:

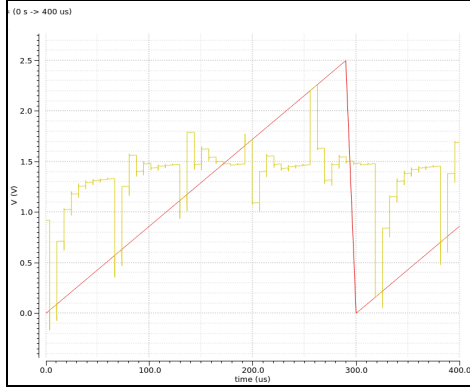


Figure 10. Sample/hold DAC input (red) and output (yellow) signal when operating within ADC.

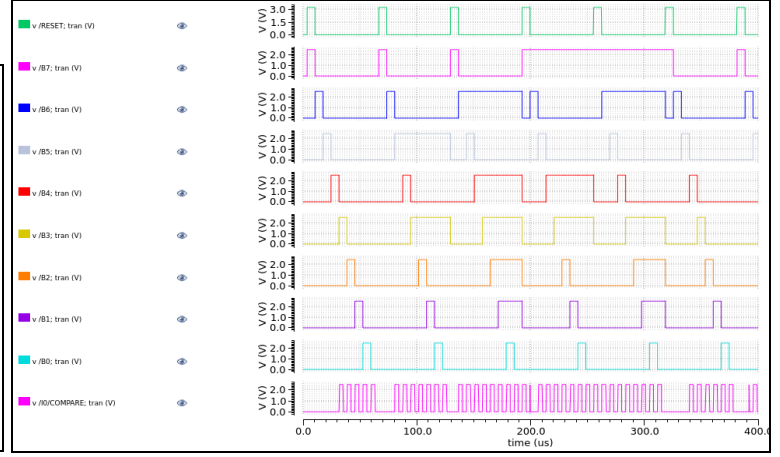


Figure 11. SAR logic operation.

Integrating all ADC subcomponents (sample/hold DAC, comparator, SAR logic) and using the testing parameters described in the previous paragraph, we realized a SAR ADC that successfully approximates the analog input within ~ 200 mV (while not the most accurate, this suits our purpose of differentiating between four distinct voltages evenly spaced within 0-2.5 V). This was determined with the following calculation:

$$V_{out} = \frac{(\sum_{n=0}^7 2^n B_n) V_{ref}}{255}$$

Equation 2. Digital output calculation. This calculation was further normalized for various input voltage rise/fall times in the Cadence calculator.

Overall ADC results are depicted in Figure 14 below:

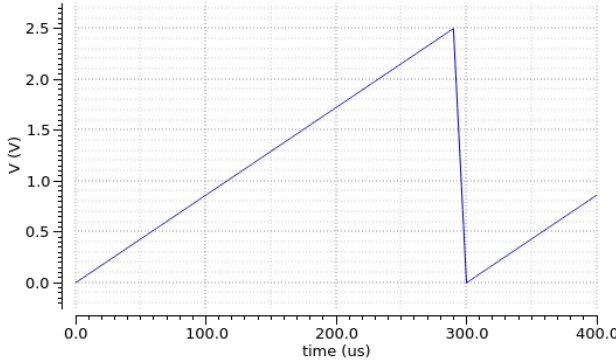


Figure 12. SAR ADC input.

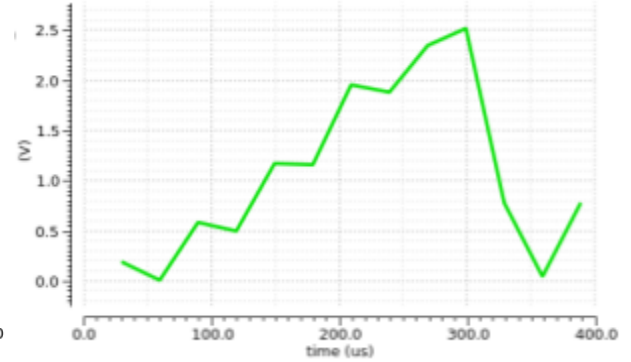


Figure 13. SAR ADC output.

Top Level

Design & Verification

At the top level, we were not able to fully verify functionality due to ECE Linux constraints. We could not run the complete system—even a 4×4 array with all CDS buffers and SAR ADCs—within a reasonable time. However, we did complete a long (~ 3 -hour) simulation where we tested the 32×32 pixel array together with the CDS stage, and the results matched our expectations: CDS reduced noise and produced a smoother, more consistent output across the array.

Even though we could not simulate the entire integrated top-level design, we thoroughly verified each individual component (pixel, CDS, ADC, and supporting biasing). Based on this, we believe full top-level functionality is primarily a matter of tuning device sizing and bias currents rather than correcting major architectural issues. To gain confidence in the integration, we also performed a dummy top-level connectivity test. In this test, we generated input pulses that emulated pixel outputs and drove four representative “pixel” voltage levels—0.5 V, 1.2 V, 2.1 V, and 3.0 V—into the CDS. With a CDS noise offset of 0.25 V, the CDS output shifted the signal as expected, and the ADC produced four distinct digital codes: 00, 01, 10, and 11. This confirms correct connectivity and signal flow end-to-end (pixel-like stimulus → CDS → ADC). Refer to the figures 14, 15 and 16: Top shows the complete top-level schematic, and Dummy shows the connectivity simulation and the expected digital outputs.

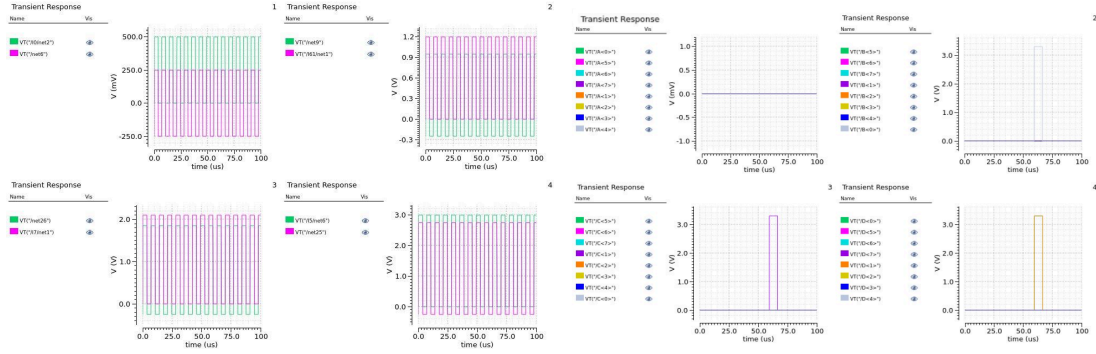


Figure 14 & 15. Top Level Dummy Simulations

Finally, at a theoretical level, the full top-level verification would follow a straightforward row-scan procedure repeated 32 times: reset Row0 → read the row signal → pass into CDS → convert with ADC, then move to Row1 and continue through Row31. To implement this, we would pulse the pixel reset for each row, then immediately pulse the CDS reset so CDS captures the reset reference level. Next, we would assert TX and ROW to transfer the pixel signal and enable row readout while TX is active. The CDS calc signal is also pulsed high during row readings. Since our SAR ADC has a 9-cycle conversion latency, we would hold the ROW signal high for 9 cycles before advancing to the next row. We would also align ADC reset with each row read so every conversion starts from a known state. This timing plan provides a clean method to validate full-frame functionality once simulation resources allow.

Layout

The layout prioritizes pixel density to maximize image sensor resolution within a fixed die area. Achieving this density requires a highly symmetrical design strategy, where adjacent pixel cells are mirrored or rotated to facilitate the sharing of supply rails. This approach mirrors the design architecture of SRAM bitcell arrays, which similarly utilize symmetry and shared power routing to minimize footprint and optimize performance across a dense grid.

In terms of area usage, much of the area overhead in design is a result of the large analog subblocks. Each pixel cell requires two poly resistors, which constitute a large area. Even more are the CDS buffers, in which 32 is needed for each sensor column. Each CDS buffer uses two 40db gain OTAs connected to an equally large 8-bit SAR ADC. The SAR ADC is large due to its well-matched capacitive DAC needed for sampling and holding.

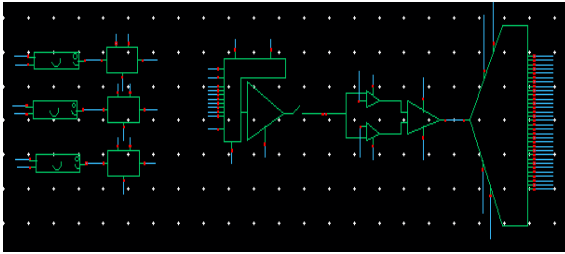


Figure 16. Top Level Schematic

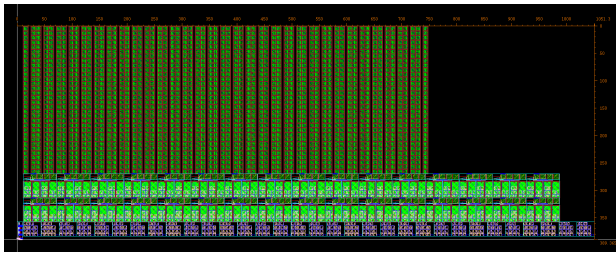


Figure 17. Top Level Layout

Results

Block	Spec	Target	Achieved
Pixel	Output voltage range	[0V, 3.3V]	[0V, ~2.5V]
Level Shifter	Power (32-bit)	10 μ W	7.706 μ W
CDS Buffer	Power	1mW	1.72mW
	Input referred noise	100 μ V rms	63 μ V rms
ADC	Resolution	8-bit	8-bit
	Clock speed	100kHz	143 kHz
	Acquisition time	80 μ s	63 μ s
	Accuracy	$\pm$ 100 mV	$\pm$ 200 mV

Discussion

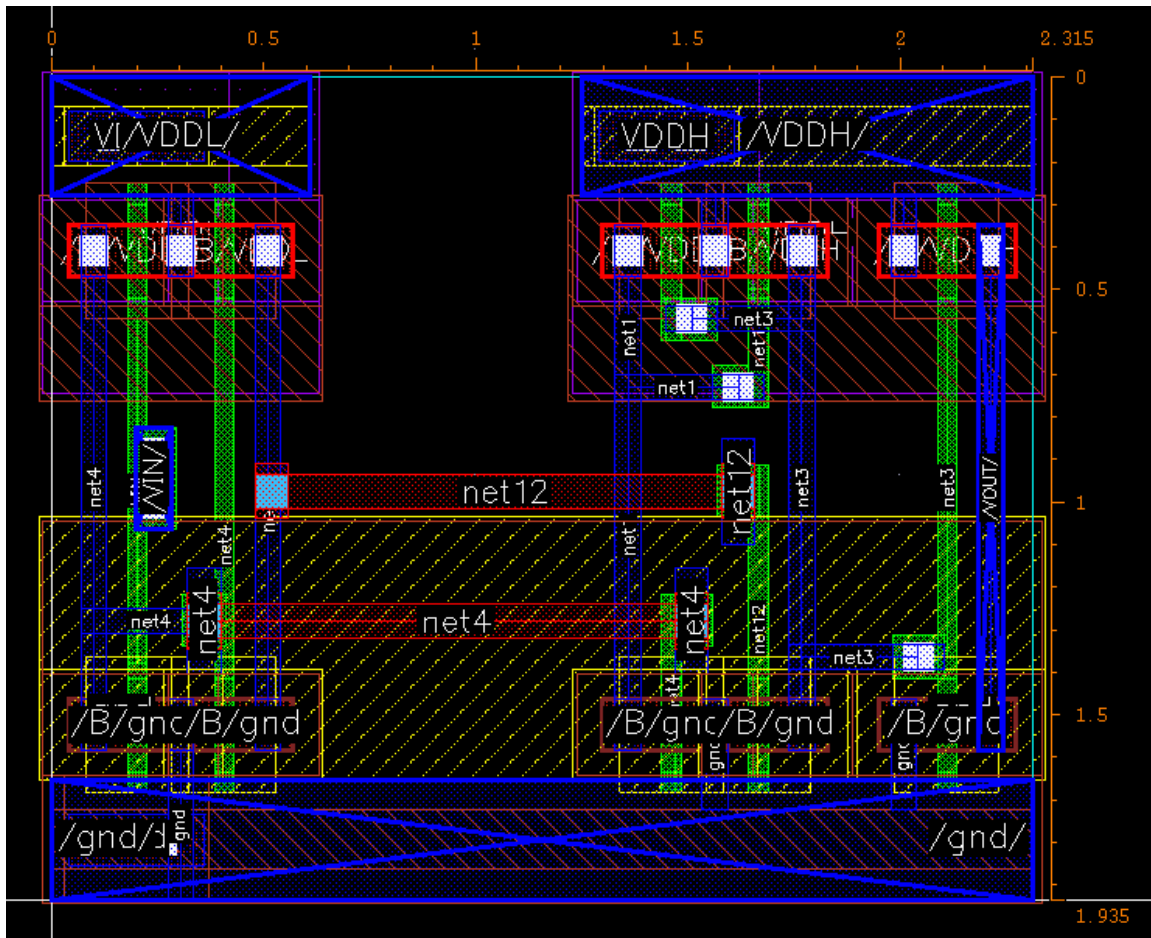
The main challenge of this project was integrating the 4T pixel, CDS buffer, and SAR ADC under realistic timing and loading conditions. For the pixel, the main difficulty was building a realistic testbench and setting the correct source-follower bias while meeting tight pulse timing requirements (reset, TX, and row select). The pixel output did not span the full 0-3.3 V range due to approximately two V_{th} drops in the readout path, and small voltage drops were observed from parasitic capacitances and charge redistribution (per Professor Molnar). The CDS buffer required balancing speed, noise, and linearity at low supply voltage. The op amp design could have been much more optimized to take these into account, but issues such as charge injection and clock feedthrough from switches, along with OTA noise, drove the choice of a moderate-gain ($\sim$ 40 dB), high-UGBW ($\sim$ 250 MHz) amplifier to ensure fast settling. The SAR ADC was primarily limited by timing, as the use of negative-edge-triggered D flip-flops made bit sequencing sensitive to clock alignment, requiring careful timing refinement for reliable conversion. Overall, integration revealed that unit-level verification is insufficient, and achieving robust system operation required iterative adjustment of biasing, device sizing, and timing to accommodate real loading and parasitic effects.

The biggest overall lesson was that top-level integration still requires iteration. When blocks are connected, real column loads and settling constraints can require larger devices for drive strength, and bias conditions often need to be revisited. We also learned to expect “weird” but real behaviors (limited swing from V_{th} drops, transient drops from parasitics, and switching artifacts in CDS), and that designing robust timing and biasing strategies is just as important as getting the schematic “functionally correct” in isolation.

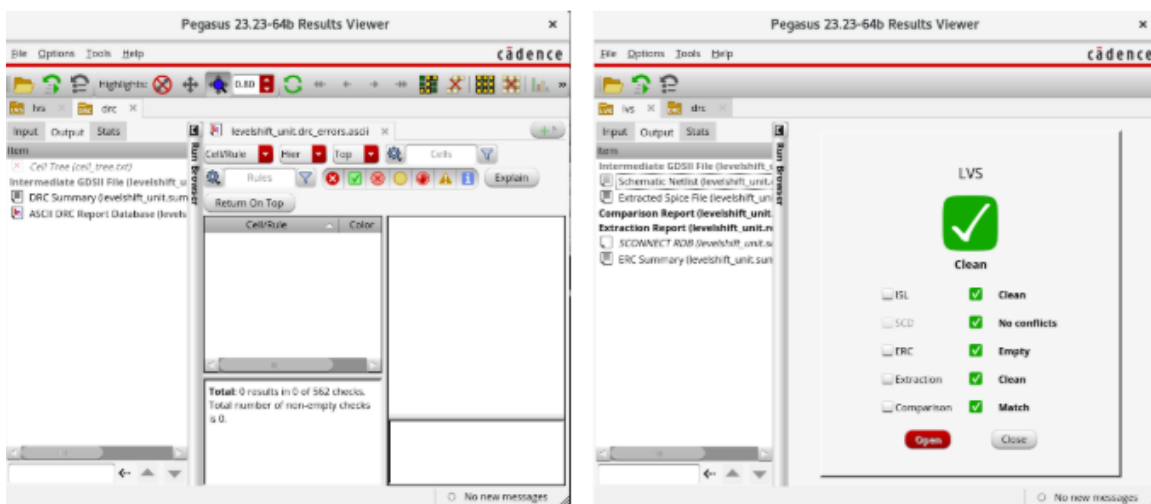
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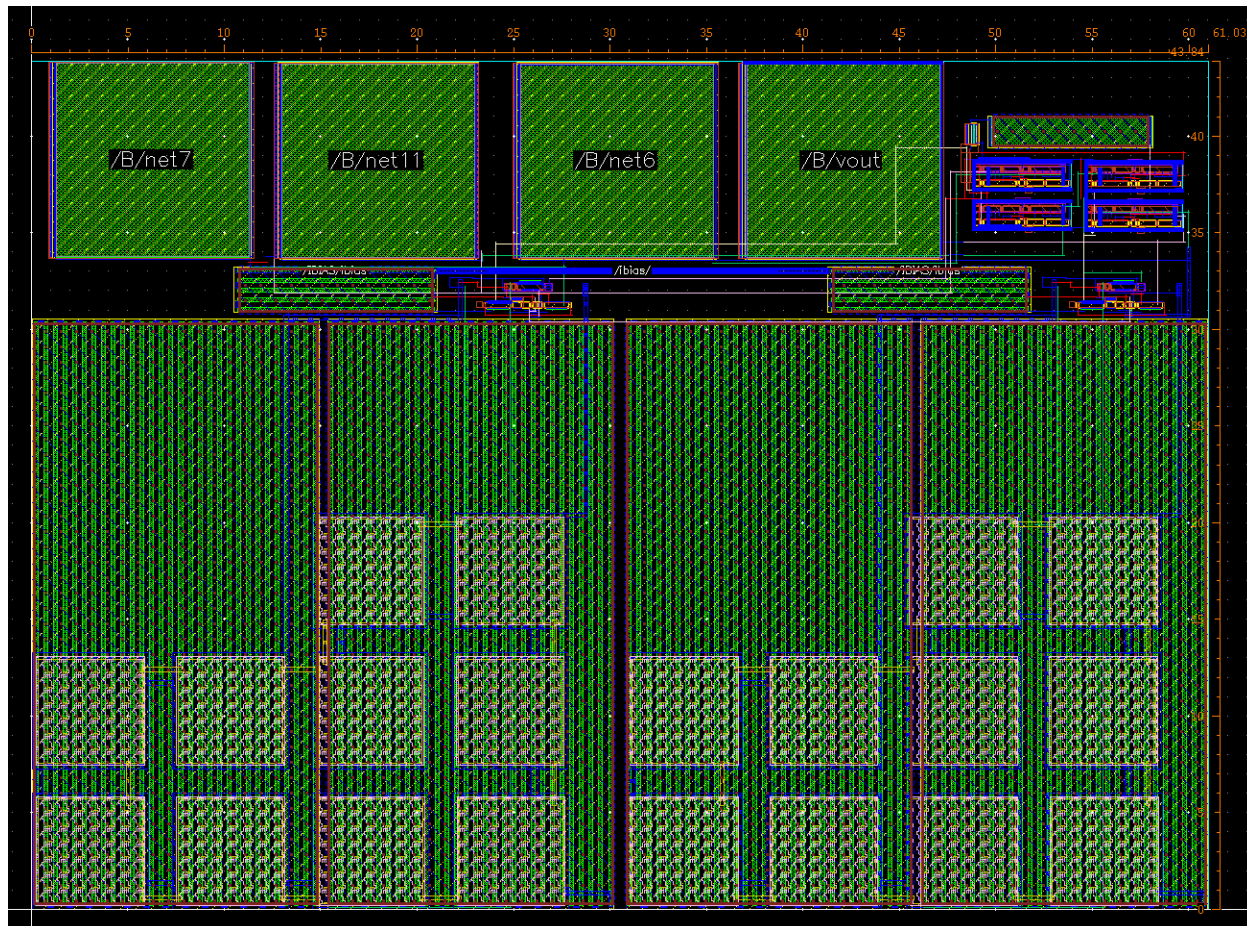
Appendix A



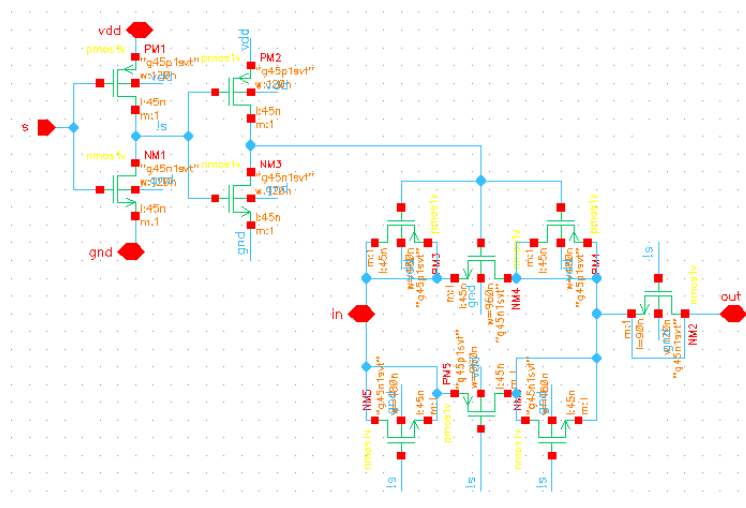
Level Shifter 1-bit Cell Layout (2.315 μm x 1.935 μm)



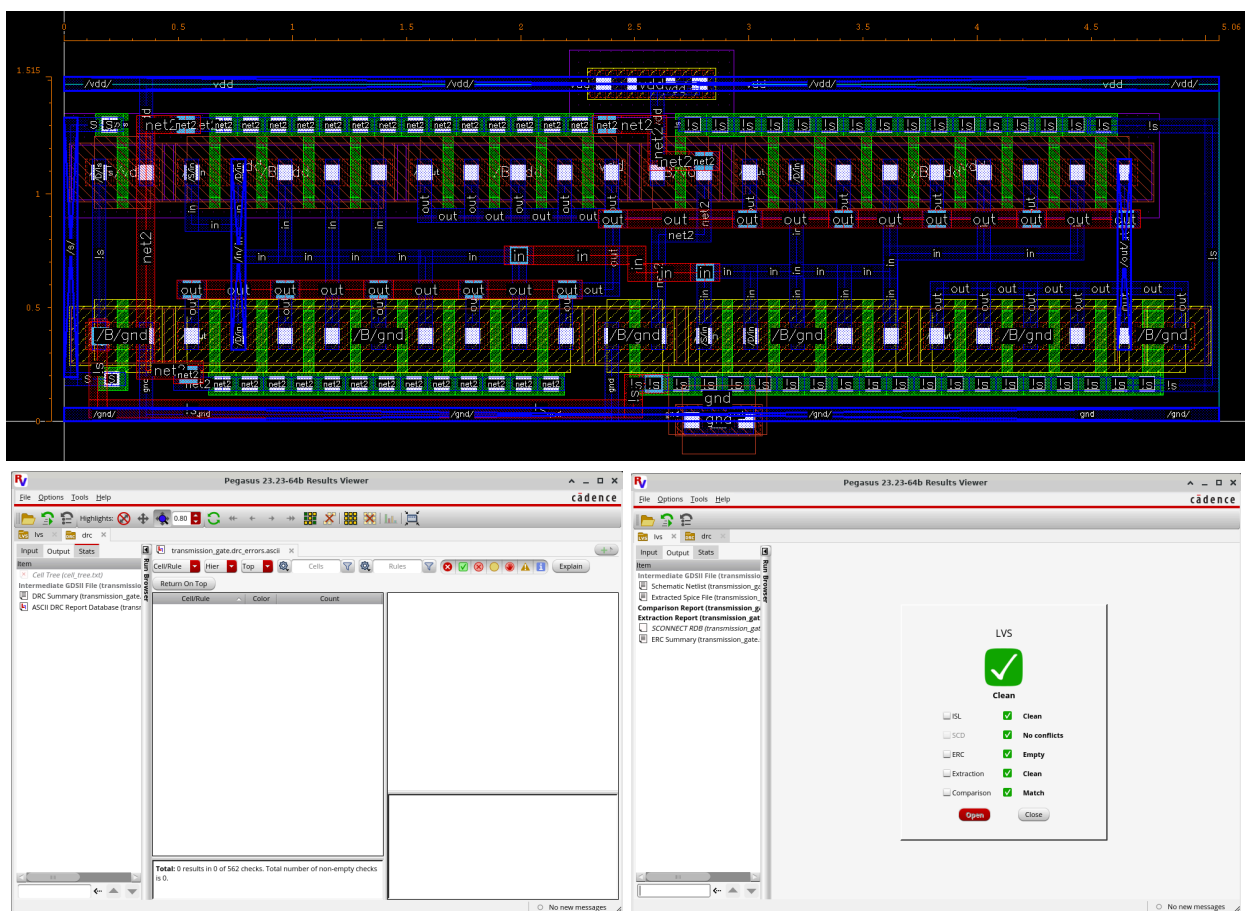
DRC and LVS Clean



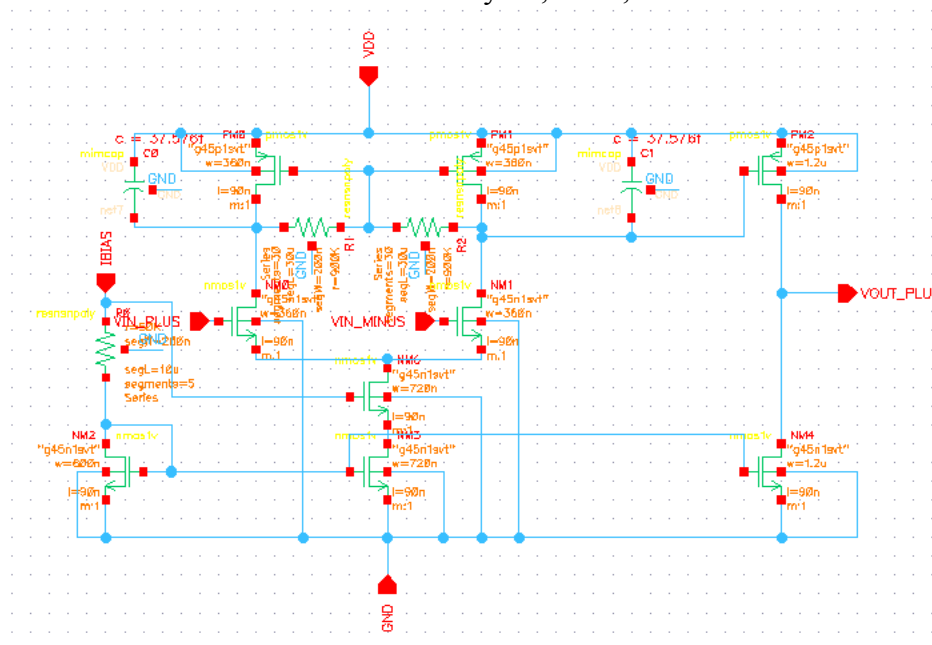
CDS Buffer Layout



Transmission Gate Schematic



Transmission Gate Layout, DRC, and LVS



Op Amp Schematic

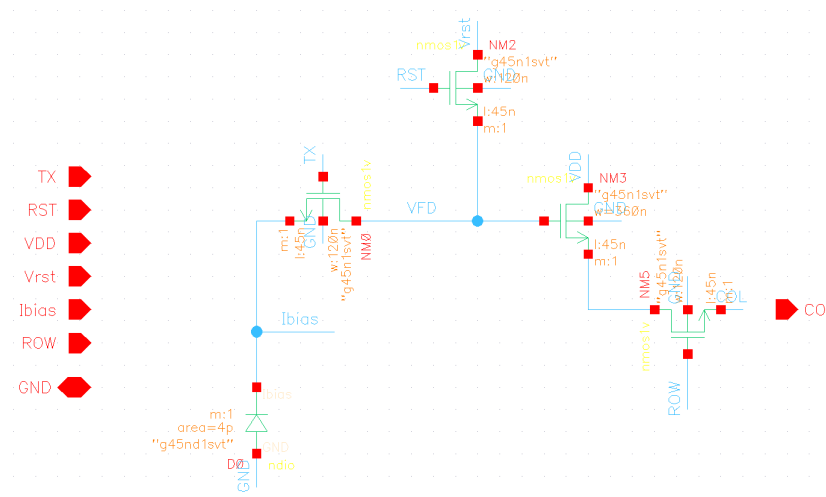
Device	Param	Noise Contribution	% Of Total
/IO/NM0	fn	1.82006e-15	39.70
/IO/I21/NM2	fn	7.45788e-16	16.27
/IO/I24/NM2	fn	7.45276e-16	16.26
/IO/I21/NM3	fn	4.77392e-16	10.41
/IO/I24/NM3	fn	4.7706e-16	10.41
/IO/I23/PM1	fn	9.67389e-17	2.11
/IO/I22/PM1	fn	9.65523e-17	2.11
/IO/I24/NM0	fn	6.14661e-17	1.34
/IO/I21/NM0	fn	6.12651e-17	1.34
/IO/NM0	id	2.85558e-19	0.01
/IO/I21/NM3	id	2.51275e-19	0.01
/IO/I24/NM3	id	2.5113e-19	0.01
/IO/I12/PM1	fn	2.21693e-19	0.00
/IO/I10/PM1	fn	1.48933e-19	0.00
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/IO/I24/NM2	id	1.40674e-19	0.00
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/IO/I23/NM2	igb	8.77004e-20	0.00
/IO/I23/PM1	id	8.26391e-20	0.00
/IO/I22/PM1	id	8.25795e-20	0.00

Integrated Noise Summary (in A²) Sorted By Noise Contributors
Total Summarized Noise = 4.58412e-15
Total Input Referred Noise = 4.81324e-08
The above noise summary info is for noise data

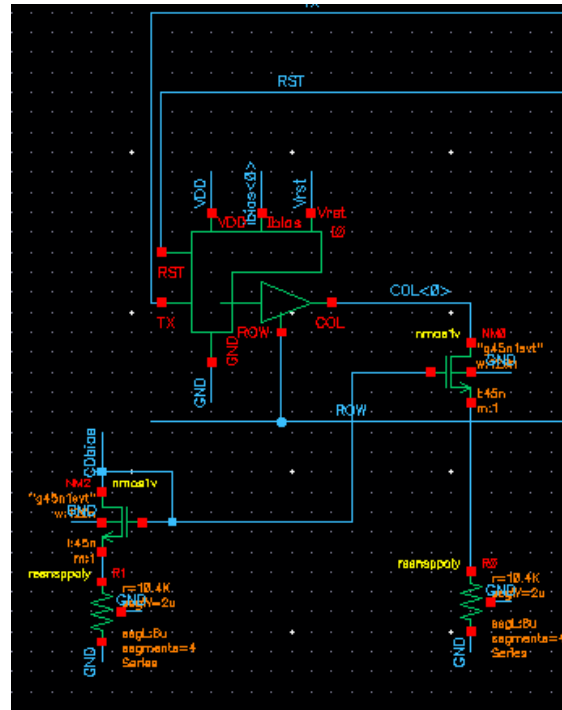
Integrated Noise Summary (in V²) Sorted By Noise Contributors
Total Summarized Noise = 4.23116e-12
Total Input Referred Noise = 4.09901e+09
The above noise summary info is for noise data

CDS Buffer Noise Summary

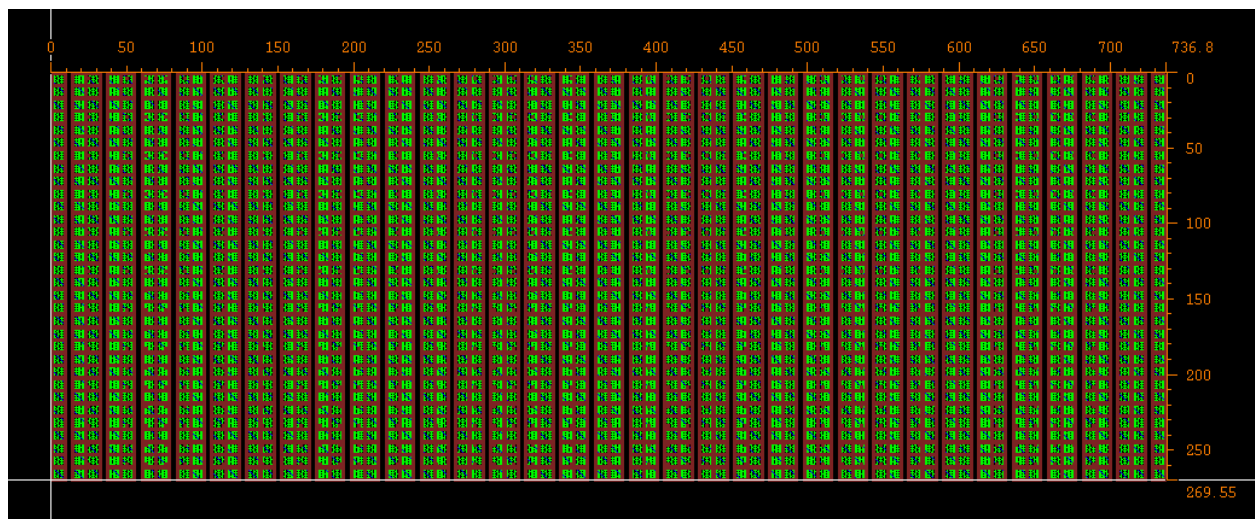
Appendix C



4T Pixel Schematic

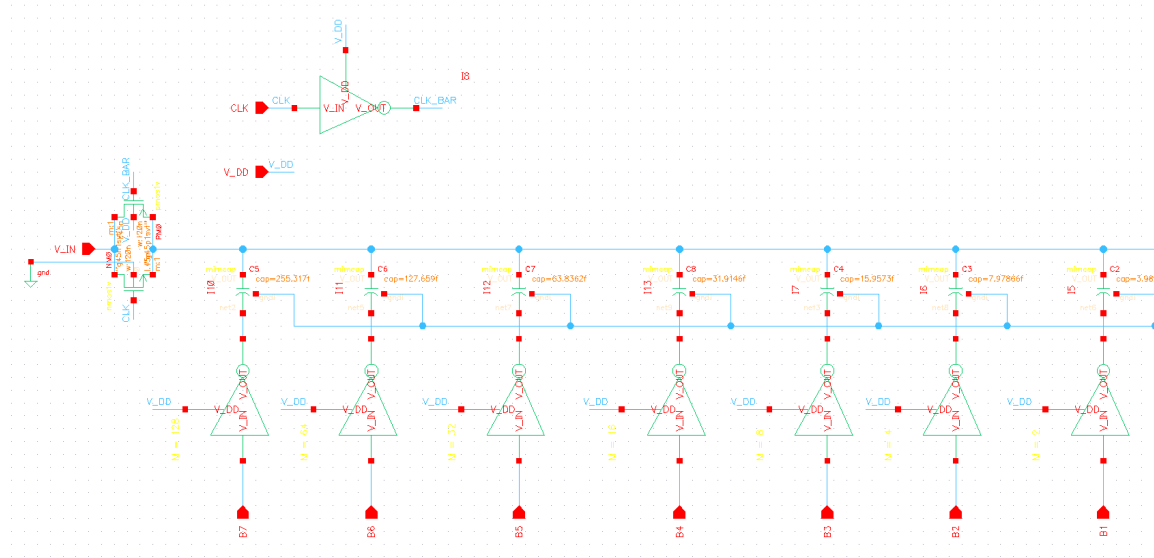


Pixel Column Schematic



Pixel Layout

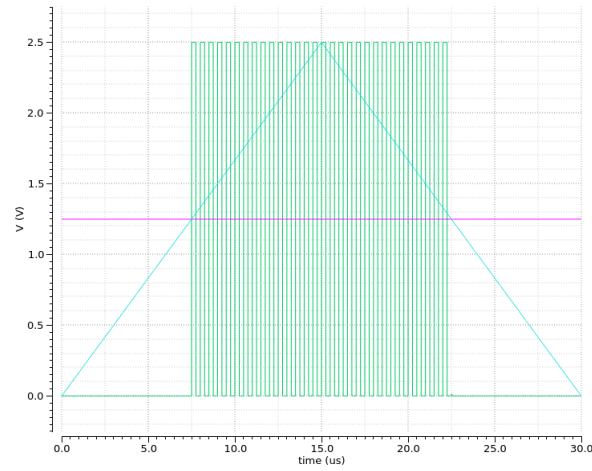
Appendix D



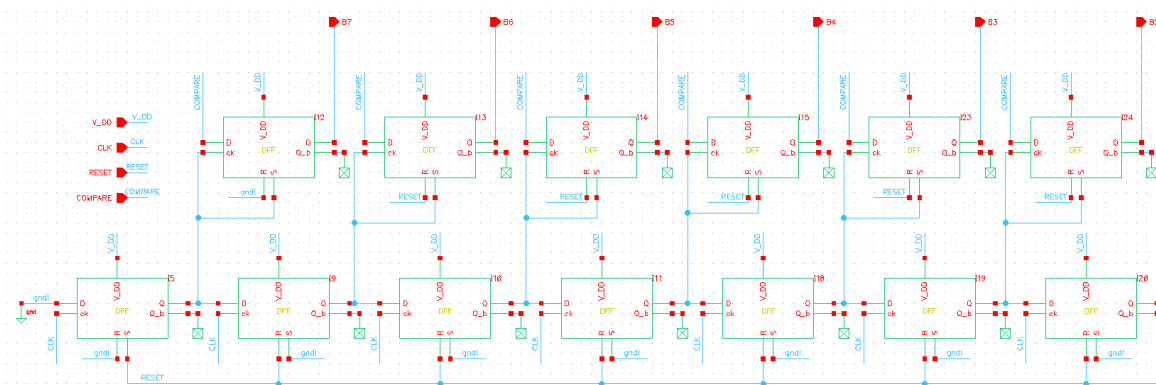
Sample/hold DAC schematic.

$$\frac{dV}{dt} = -\left(\frac{I_{bias}}{2} \pm \Delta I\right) / C_p$$

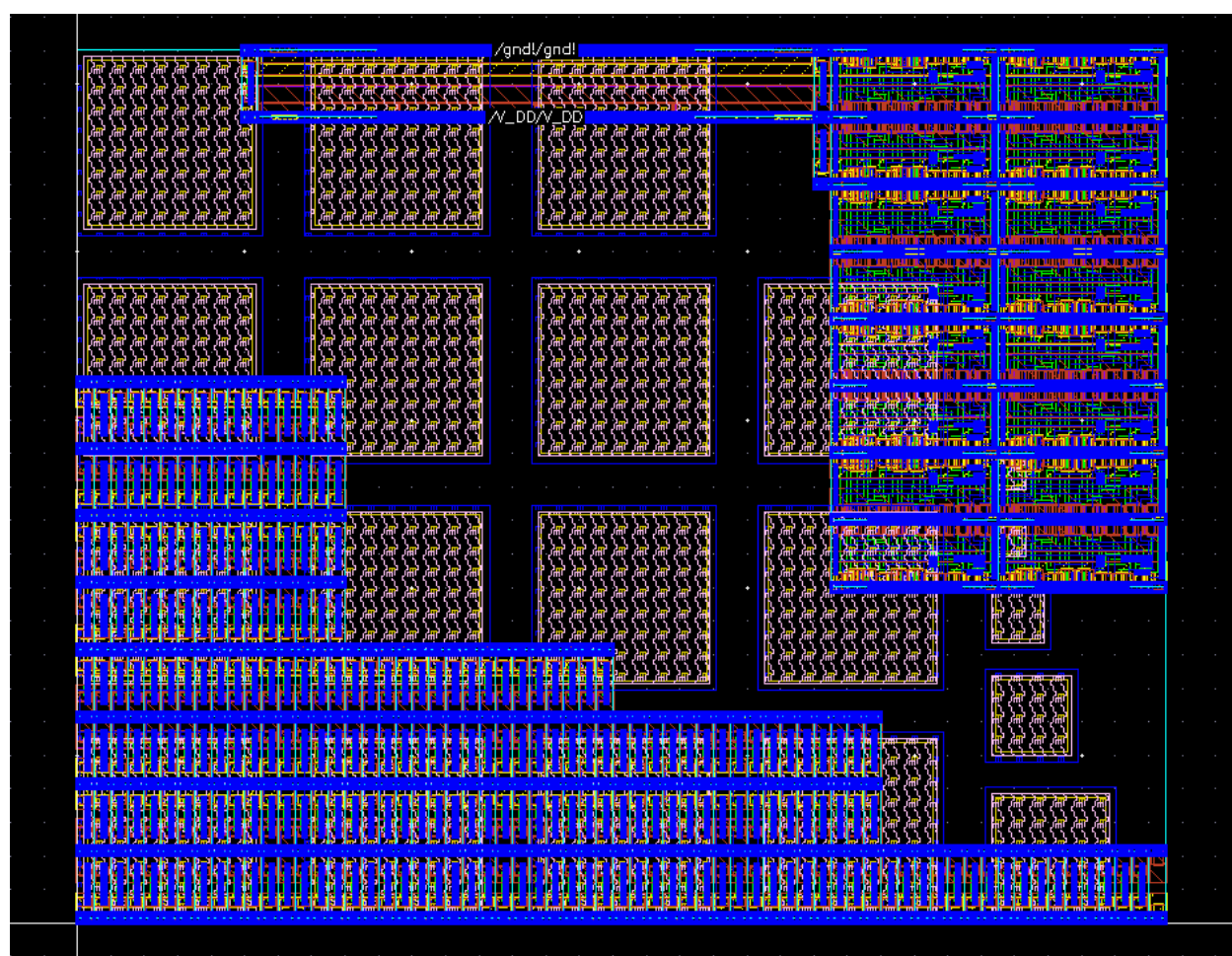
Slew down rate of internal nodes in comparator.



Comparator output (green), input (pink), and reference voltage (blue).



SAR digital logic schematic.



8-bit SAR Layout